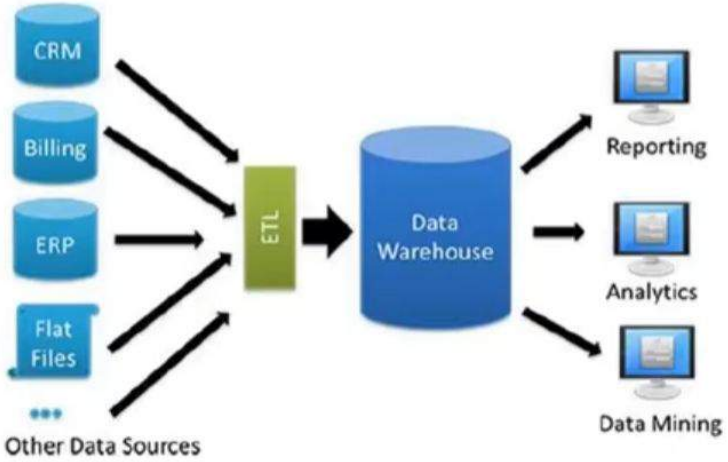
What is HIVE ?

- Hive is a data warehouse infrastructure that is built on top of Hadoop
- Hive provides a mechanism to project structure onto the data and query the data using SQL-like language called HiveQL
- Hive uses Map Reduce and HDFS for Processing and storage/ retrieval of data



HIVE Advantages

The advantages of using Hive are:

- It can be used as an ETL.
- Provides capability of querying and analysis.
- Can handle large data sets.
- SQL(filters, joins, group by) on top of Map and Reduce.

Where not to use HIVE

Hive should not be used:

- If the data does not cross GB's.
- If we don't find schema or bringing in schema is difficult or not possible on the data at hand.
- If we need response in seconds and for low latency applications.
- If RDBMS can solve, don't invest time in Hive.

HIVE Features

Similarity with SQL

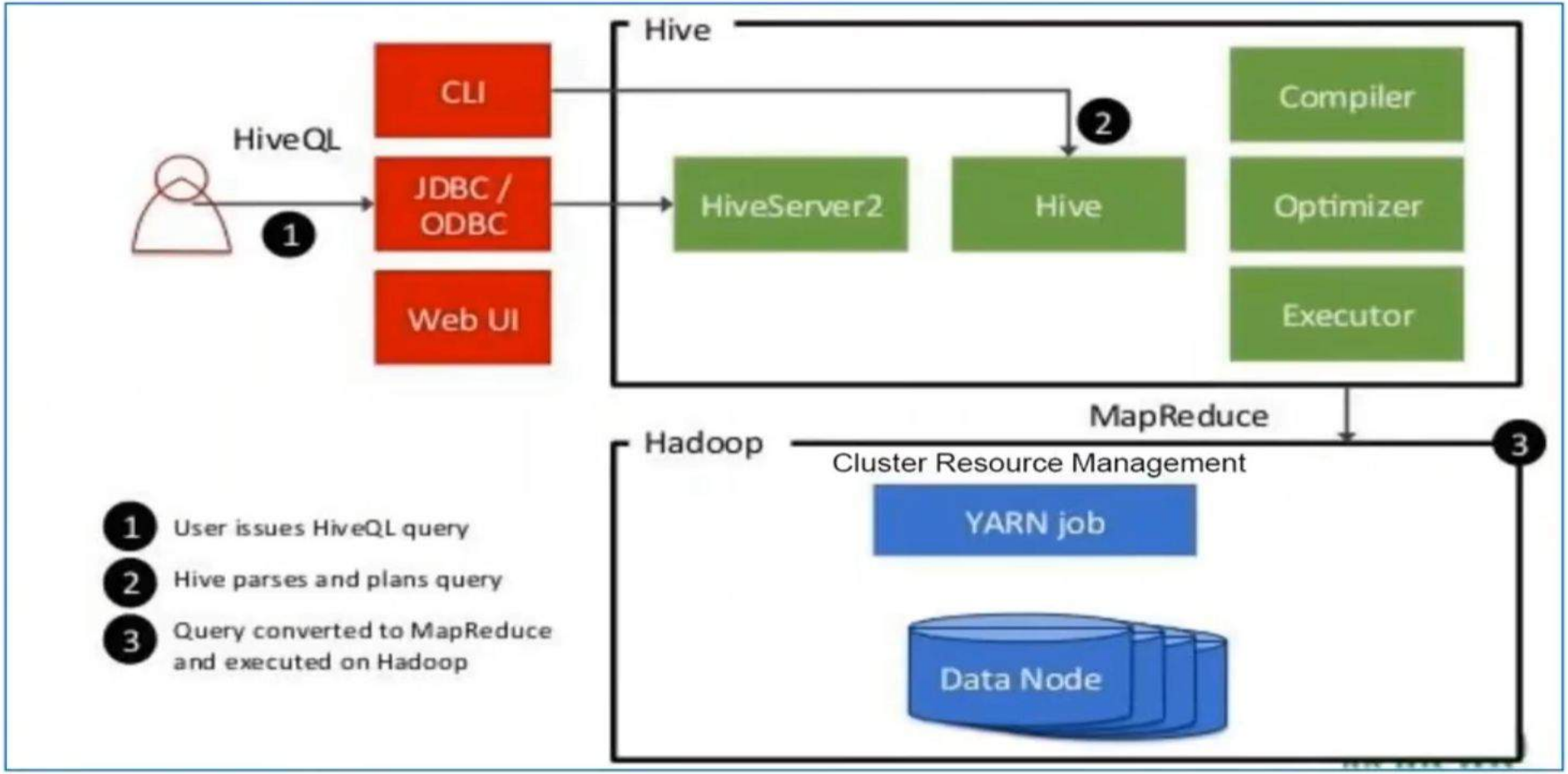
- Hive is very similar to RDBMS SQL like queries
- HiveQL is based on SQL-92 framework and it is safe to say that its functions as SQL for Hadoop DFS.

Difference from SQL

- The major difference is that a Hive query executes on a Hadoop infrastructure rather than a traditional database.
- This allows Hive to scale to handle huge data sets - data sets so large that high-end, expensive, traditional databases would fail.
- The internal execution of a Hive query is via a series of automatically generated Map Reduce jobs

HIVE Vs MapReduce

All Hive queries will be converted to Map Reduce Job for execution, so why can't we write Map Reduce our selves?

- Understanding the internals of Hadoop Framework is must to write Map Reduce Jobs.
- Engineer having SQL knowledge can quickly write Hive Scripts and get the results.

HIVE Architecture